

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – NUMERICAL HACKATHON

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	Assessment:	Assessment Tool 3 – HACKATHON
Weightage:	30 % of CIA	Total Marks:	100
CO3 Statement:	Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks. (BTL 3)		
Student Name:	I.V. Growtham Reddy	Reg. No.:	192372158
Section / Batch:	CSE AI / 2023	Date:	12/08/2026

Code of Conduct

[192372158]

I, I.V. Growtham Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Growtham

Instructions

- Duration: 60 minutes
- Number of questions : 10

Faculty In-charge [Terrance Frederick Fernandez]	Course Coordinator [Terrance Frederick Fernandez]	Head of Department
Signature: <u>[Signature]</u>	Signature: <u>[Signature]</u>	Signature: _____
Date: <u>12/08/26</u>	Date: <u>12/08/26</u>	Date: _____

95
100 ✓

1. Given)

Data packet size = 500 bytes.

Tunnel overhead = 20 bytes

Additional overhead of 500-byte = 20 bytes,,

2. Bandwidth = 20 MHz

64-level Quadrature Amplitude Modulation = 6 bits/symbol
[64-QAM] Each symbol carries 6 bits $\Rightarrow \log_2(64) = 6$ Data Rate = Bandwidth $\times$ Bits per Symbol

$$= 20 \times 10^6 \times 6$$

$$= 120 \times 10^6 \text{ bits/s}$$

$$= 120 \text{ Mbps}$$

The maximum theoretical data rate is 120 Mbps.

3. No. of entries = 50

memory per entry = 32 bytes

Total memory = $50 \times 32 = 1600$ bytesConverting to KB : $1600 \div 1024 \Rightarrow 1.56 \text{ KB}$

proactive - Routes are maintained continuously before they are needed.

Reactive - Routes are discovered only when they are needed

4. Speed = 10 m/s

Update Interval = 2 seconds

Distance = Speed $\times$ Time

$$= 10 \times 2$$

$$= 20 \text{ meters,,}$$

5] Given) 5 Intermediate nodes

$$\text{Total} = 5 + 1 \text{ (originator)}$$

$$\text{Total packets} = 6 //$$

6] Given) Subnet mask = 255.255.255.240

$$\text{Last octet} : 240$$

$$\text{Host bits} = 4$$

$$\text{Total addresses} = 2^4 = 16$$

$$\text{Usable host addresses} = 16 - 2 = 14 \text{ mobile nodes} //$$

7] Given)

$$\text{channel bandwidth} = 20 \text{ MHz}$$

$$\text{Symbol rate} = 1 \text{ Msps [Mega Symbols per Second]}$$

$$\text{Spectral Efficiency} = \frac{\text{Symbol rate}}{\text{Bandwidth}}$$

$$= \frac{1 \times 10^6}{20 \times 10^6} = 0.05 \text{ Symbols/s/Hz}$$

8] Given)

$$\text{No. of nodes} = 50$$

$$r = 250 \text{ m}$$

$$\text{Area} = 1 \text{ Km}^2 = 10^6 \text{ m}^2$$

$$\text{Network Density} = \frac{\text{No. of nodes} \times \pi r^2}{\text{Area}}$$

$$= \frac{50 \times \pi \times (250)^2}{10^6}$$

$$= \frac{50 \times 3.1416 \times 62500}{10^6}$$

$$\approx \frac{9817500}{10^6}$$

$$\approx 9.82 \text{ (nodes per transmission area)}$$

Given)

$$V = 30 \text{ m/s}$$

$$C = 3 \times 10^8 \text{ m/s}$$

$$f_c = 900 \times 10^6 \text{ Hz}$$

$$f_d = \frac{V}{C} \times f_c$$

$$= \frac{30}{3 \times 10^8} \times 900 \times 10^6$$

$$= 90 \text{ Hz,,}$$

⇒ Doppler shift is change in frequency of a wireless signal caused by relative movement between the transmitter and receiver

• Moving towards → frequency increases

• Moving away → frequency decreases

10] Given)

$$\text{Data rate} = 54 \text{ Mbps}$$

$$\text{Packet size} = 1500 \text{ bytes}$$

Converting bytes to bits : $1500 \times 8 = 12000 \text{ bits}$

$$\text{Transmission time } T = \frac{\text{Packet Size (bits)}}{\text{Data rate (bits/s)}}$$

$$= \frac{12000}{54 \times 10^6}$$

$$= 0.0002222 \text{ s}$$

$$T \approx 222.2 \mu\text{s}$$

Transmission Time [T] = 222.2 microseconds (μs) per packet.